
UNIT 14 CANONICAL CORRELATION ANALYSIS

Structure

- 14.0 Objectives
- 14.1 Introduction
- 14.2 Canonical Correlation Analysis (CCA): Concept and Meaning
- 14.3 Assumptions of Canonical Correlation
- 14.4 Canonical Correlation Analysis as Generalization of the Multiple Regression Analysis
- 14.5 Steps and Procedure Involved in Computation of CCA Results
- 14.6 Illustration of CCA
- 14.7 Interpretation of CCA Results
- 14.8 Limitations of Canonical Correlation
- 14.9 Let Us Sum Up
- 14.10 Key Words
- 14.11 Some Useful Books
- 14.12 Answers or Hints to Check Your Progress
- 14.13 Exercises

14.0 OBJECTIVES

After going through this unit, you will be able to:

- explain the concept of canonical correlation analysis;
- state the similarity and difference between multiple regression and canonical correlation;
- discuss the steps and procedure involved in canonical correlation;
- elucidate how to interpret the results of canonical correlation;
- point out the limitations of canonical correlation analysis.

14.1 INTRODUCTION

The conventional wisdom that the economic agents are rational and are guided by self-interest in decision making is being questioned as it ignores the psychological and social factors influencing decision making process. The research findings from many disciplines like neuroscience, cognitive science, psychology, behavioral economics, sociology, anthropology etc. indicate that the decisions made by the individuals in many aspects of development (like savings, investment, energy consumption, health etc.) are influenced by social contexts, local social networks, cultural factors, social norms and shared mental models etc. (World Development Report, 2015). Hence, inter-disciplinary perspective is being recognized as research approach to analyze human behavior so as to improve the predictive power of economics. Canonical

correlation analysis is a powerful analytical tool to analyze the association between two sets of variables belonging to different disciplines. This method measures the strength of association between the two sets of variables. An attempt is made in this method to concentrate a high-dimensional relationship between two sets of variables into a few pairs of canonical variables. Hence, in this unit, we shall throw light on various issues relating to canonical correlation like concept and meaning of canonical correlation, its similarity and differences with multiple regression, procedure involved in the analysis of canonical correlation, and its advantages and limitations.

14.2 CANONICAL CORRELATION ANALYSIS (CCA): CONCEPT AND MEANING

Canonical correlation analysis is a multivariate statistical model used to study the interrelationships among sets of multiple dependent variables and multiple independent variables. This technique is distinct from the multiple regression model in the sense that multiple regression predicts a single dependent variable from a set of multiple independent variables whereas canonical correlation simultaneously predicts multiple dependent variables from multiple independent variables.

Let us understand the concept with an example. As a student of economics, you may like to know the association between economic inequality X^* and political instability Y^* . The economic inequality can be measured by five variables i.e. (i) the division of farmland (X_1), (ii) the gini coefficient (X_2), (iii) the percentage of tenant farmers (X_3), (iv) the gross national product (X_4), and (v) the percentage of farmers (X_5). Similarly the political instability can be measured by four variables (indicators) i.e. (i) the instability of leadership (Y_1), (ii) the level of internal group violence (Y_2), (iii) the occurrence of internal war (Y_3), (iv) stability of democracy (Y_4). These two theoretical concepts X^* and Y^* can be called two sets of variables or canonical variables. These can be shown in the following figure:

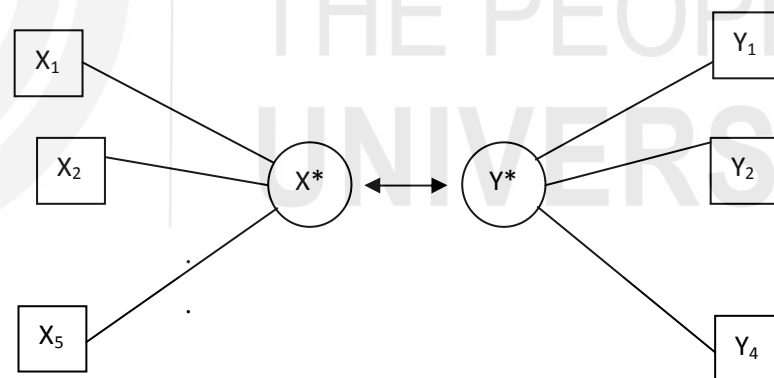


Fig. 14.1: Canonical Correlation

Source: The sage encyclopedia of Social Sciences Research Methods vol. 1 (2004) page no. 83.

The first canonical variable X^* is measured by five variables ($P = 5$) and can be considered as a linear combination (a weighted sum) of these X variables. The second canonical variable Y^* is a linear combination of the $q = 4$ indicators, Y_1 to Y_4 . The double side curved arrow indicates that the question of casualty remains open.

The purpose of canonical correlation analysis is to find the correlation between a linear combination of the variables in one set and a linear combination of the variables in another set. The idea behind this approach is first to determine the pair of linear combinations having the largest correlation. Next, we determine the pair of linear combinations having the largest correlation among all pairs uncorrelated with the

initially selected pair, and so on. As stated in the above para, the pairs of linear combinations are called the canonical variables and their correlations are called canonical correlations.

Thus, canonical correlation aims to (i) identify the dimensions among the dependent and independent variables, and (ii) maximize the relationship between the dimensions.

In this manner, in canonical correlation, we can distinguish three types of correlations:

- 1) Correlation between X variables, the correlation matrix is R_{xx} .
- 2) Correlation between Y variables, the correlation matrix is R_{yy} .
- 3) Correlation between X and Y variables, the correlation matrix is $R_{xy} = R'_{yx}$.

14.3 ASSUMPTIONS OF CANONICAL CORRELATION

- 1) The correlation coefficient between any two variables is based on linear relationship.
- 2) The canonical correlation is the linear relationship between the variates.
- 3) The distribution of variables is normal.
- 4) Heteroscedasticity, to the extent it decreases the correlation between variables.

14.4 CANONICAL CORRELATION ANALYSIS AS A GENERALIZATION OF MULTIPLE REGRESSION ANALYSIS

Multiple regression analysis is a multivariate technique which can predict the value of a single dependent variable from a linear function of a set of independent variables. But this is not always the case. There are real life problems, however, when interest may not center on a single dependent variable. Rather, the researcher may be interested in relationships between sets of multiple dependent and multiple independent variables. Canonical correlation analysis is a multivariate statistical model that facilitates the study of interrelationships among sets of multiple dependent variables and multiple independent variables. Whereas multiple regression predicts a single dependent variable from a set of multiple independent variables, canonical correlation simultaneously predicts multiple dependent variables from multiple independent variables. Therefore, canonical correlation analysis is said to be a generalization of multiple correlation used in multiple regression problems.

The coefficient of determination R^2 , in regression problems is the proportion of the variability in a dependent variable that is accounted for by a set of predictor variables and $R = \sqrt{R^2}$ is called the multiple correlation coefficient. The multiple correlation coefficient can also be interpreted as a measure of the maximum correlation that is attainable between the dependent variable and any linear combination of the predictor variable. Canonical correlation places the fewest restrictions on the types of data on which it operates. Because the other techniques impose more rigid restrictions, it is generally believed that the information obtained from them is of higher quality and may be presented in a more interpretable manner. For this reason, many researchers view canonical correlation as a last effort, to be used when all other higher-level techniques have been exhausted. But in situations with multiple dependent and independent variables, canonical correlation is the most appropriate and powerful multivariate technique. It has gained acceptance in many fields and represents a useful tool for multivariate analysis, particularly while considering multiple dependent variables.

14.5 STEPS AND PROCEDURE INVOLVED IN COMPUTATION OF CCA

In 1935-36, Hotelling proposed a method, known as Canonical Correlation Analysis to investigate “linear” relationship between the two sets of variates.

James Press (2005) has expressed the whole idea of this Canonical Correlation Analysis in the following words:

“The Canonical correlation” model selects weighted sums of variables from each of the two sets to form new variables in each of the sets, so that the correlation between the new variables in “different sets” is maximized while the new variables within each set are constrained to be uncorrelated with mean zero and unit variance.

We shall adhere to Press’s approach.

Let

$$\alpha: p_1 \times 1$$

and

$$\gamma: p_2 \times 1$$

be two unknown vectors to be determined such that the correlation between $\alpha'Y$ and $\gamma'Z$ be as large as possible.

So, let

$$U_1 = \alpha'Y$$

$$V_1 = \gamma'Z$$

and

$$\rho(U_1, V_1) = \text{Correlation coefficient between } U_1 \text{ and } V_1.$$

The problem of correlation coefficient now amounts to the following:

$$\text{maximize } \rho(U_1, V_1)$$

Subject to

$$\text{Var}(U_1) = \text{Var}(V_1) = 1$$

and

$$E(U_1) = E(V_1) = 0.$$

Hotelling solved this problem using the celebrated method of Lagrange Multipliers. However, we shall just state the final results. Hotelling showed that this maximization problem is equivalent to following algorithm:

Step 1: Solve for λ the equation

$$\begin{vmatrix} -\lambda\Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & -\lambda\Sigma_{22} \end{vmatrix} = 0 \quad \text{-----} \quad (A)$$

Here, $\Sigma_{11} = \text{Var}(Y)$

$$\Sigma_{22} = \text{Var}(Z)$$

$$\Sigma_{12} = \Sigma_{21} = \text{Cov}(Y, Z)$$

Let λ_1 be the largest positive root of the above equation.

Step 2: Now, solve the system of equations:

$$\begin{pmatrix} -\lambda_1 \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & -\lambda_1 \Sigma_{22} \end{pmatrix} \begin{pmatrix} \alpha \\ \gamma \end{pmatrix} = 0$$

for α and γ .

Mathematically, it is not so simple to solve the above system of linear equations. However, there is an equivalent formulation. To compute α and γ , we solve the pair of equations:

$$(\Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} - \lambda_1^2 \Sigma_{11}) \alpha = 0$$

$$(\Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12} - \lambda_1^2 \Sigma_{22}) \gamma = 0$$

Step 3: We now call these α and γ as α_1 and γ_1 respectively; and

$$U_1 = \alpha_1' Z$$

and

$$V_1 = \gamma_1' Z$$

as **First Canonical Variates**.

It will turn out that λ_1 will be the correlation coefficient between U_1 and V_1 . We, therefore, write

$$\lambda_1 = \rho(U_1, V_1)$$

and call this as **First Canonical Correlation**.

Step 4: We now proceed to the next iteration. We now define:

$$U_2 = \alpha_2' Z$$

$$V_2 = \gamma_2' Z,$$

where α_2 and γ_2 are to be determined.

$$\text{maximize } \rho(U_2, V_2)$$

subject to

$$\text{Var}(U_2) = \text{Var}(V_2) = 1$$

and

$$E(U_2) = E(V_2) = 0.$$

We repeat the above procedure to compute α_2 and γ_2 as solution of

$$\begin{pmatrix} -\lambda_2 \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & -\lambda_2 \Sigma_{22} \end{pmatrix} \begin{pmatrix} \alpha_2 \\ \gamma_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

where λ_2 is the second largest positive root of the equation (A).

Step 5: We continue this procedure until the smallest positive root.

Step 6: The result is now to be collected in a vector format:

$$U = \begin{pmatrix} U_1 \\ U_2 \\ \vdots \\ U_{p_1} \end{pmatrix} = (U_1 \dots U_{p_1})'$$

$$V = \begin{pmatrix} V_1 \\ V_2 \\ \cdot \\ \cdot \\ V_{p_1} \end{pmatrix} = (V_1 \dots V_{p_1})'$$

The elements of U and V are called **Canonical Variates** and λ_i is corresponding canonical correlation.

14.6 ILLUSTRATION OF CCA

Let us now understand the canonical correlation analysis technique with the help of an illustration.

This example is based on 416 observations collected through primary survey for the purpose of research study entitled “Assessment of Human Well-being in Delhi: Multidimensional Approach”. The researcher attempted to study the relationship between economic well-being variables and overall life satisfaction variables. One of the questions raised in this study was to examine how does economic well-being influence the overall life satisfaction of the people or more specifically, to know whether economic well-being indicators are predictive of overall life satisfaction of people.

The main characteristic of canonical analysis is the investigation of the relationship between two sets of variables. One set is the predictor set or, say, analytically, the set of independent variables. The second consists of the criteria or dependent variables. In our example, the set of economic well-being variables constitutes the set of independent variables. This set of economic well-being variables consist of 5 indicators of economic well-being:

- 1) Annual Income
- 2) Movable Assets
- 3) Fixed Assets
- 4) Employment Status
- 5) Educational Attainment

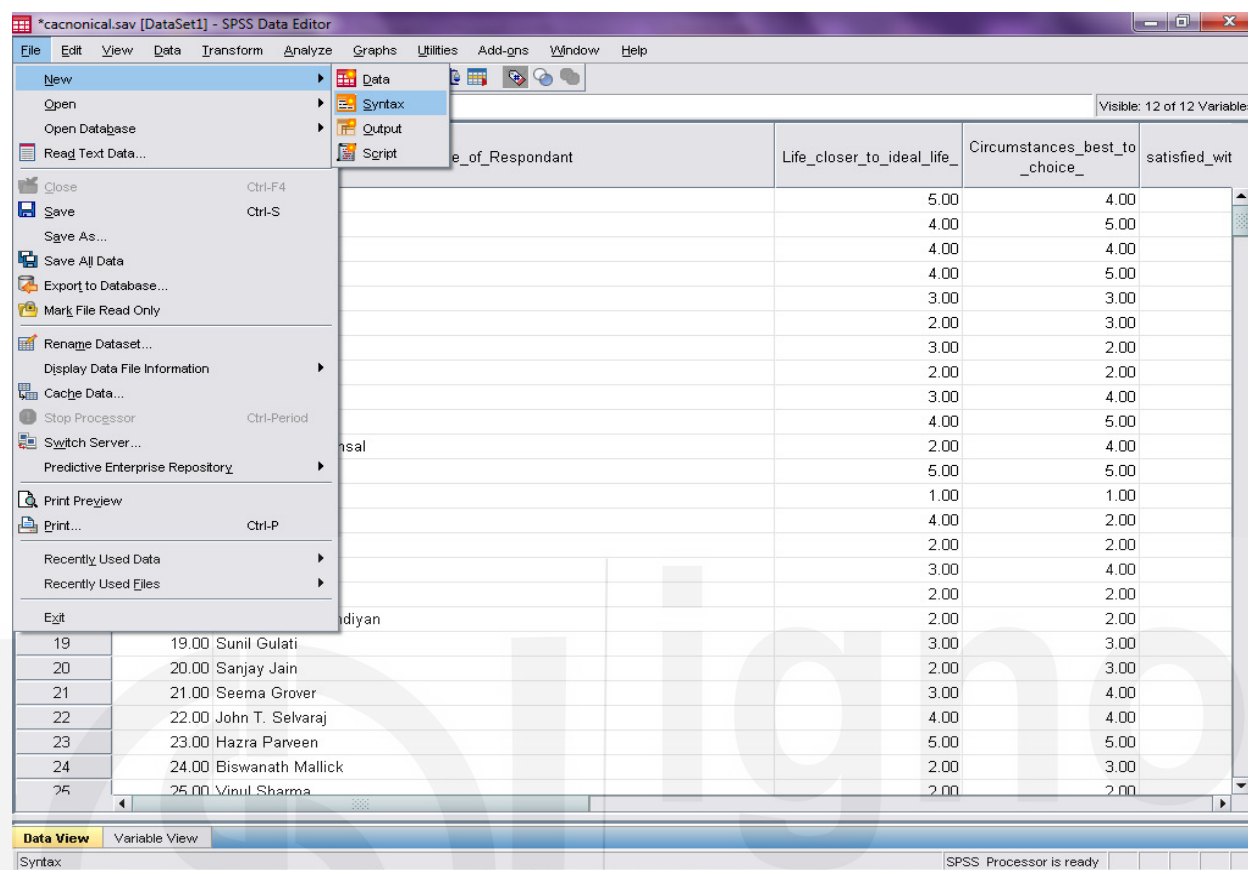
Next, the set of criteria variables constitute the overall life satisfaction indicators:

- 1) My life closer to my ideal life
- 2) Circumstances of my life are best to my choice
- 3) I am satisfied with my life
- 4) I have achieved the things in my life I aspired
- 5) I would not prefer to make any drastic change in my rest of life till I survive

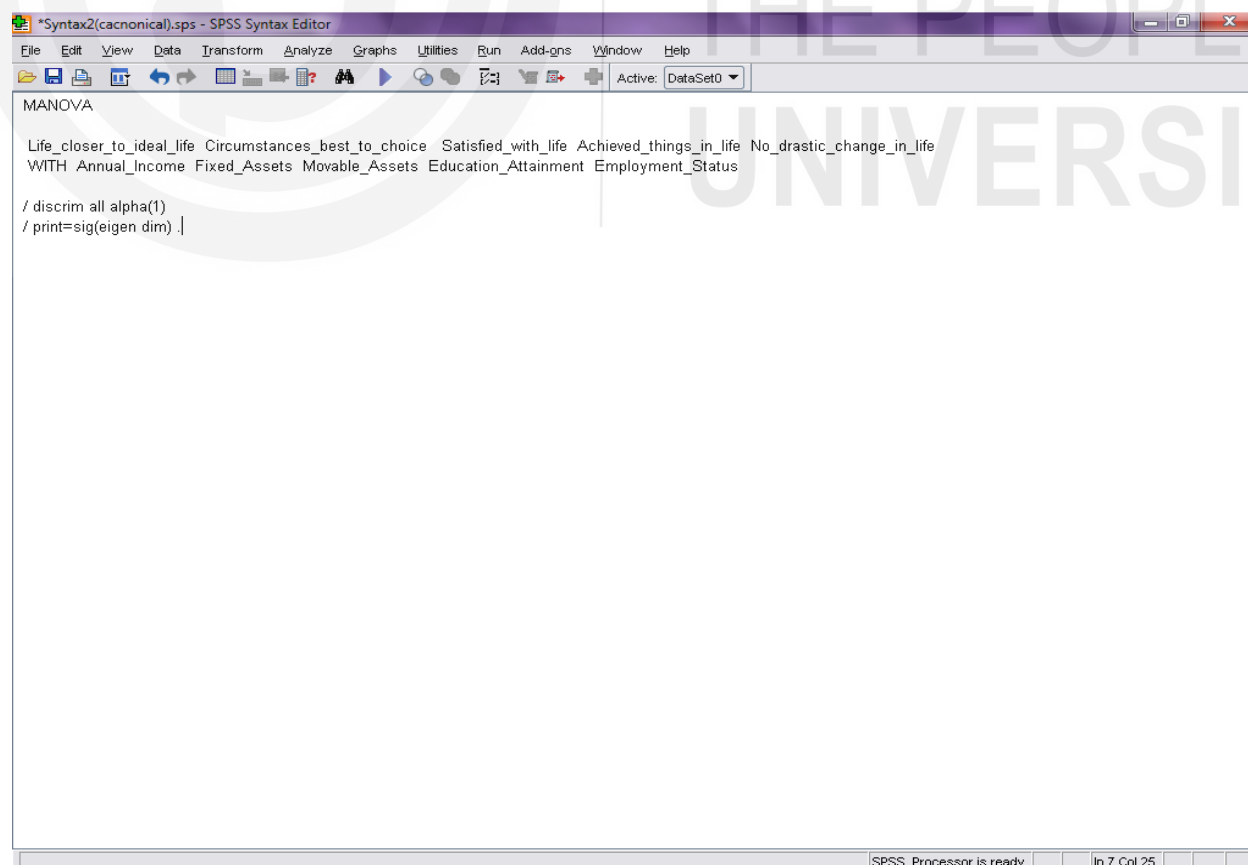
The data on these two sets of variables was collected through primary survey for the above said study.

Let us now learn to run the Canonical Correlation Analysis for our example using SPSS

Step 1: Click file, new and syntax sequence.

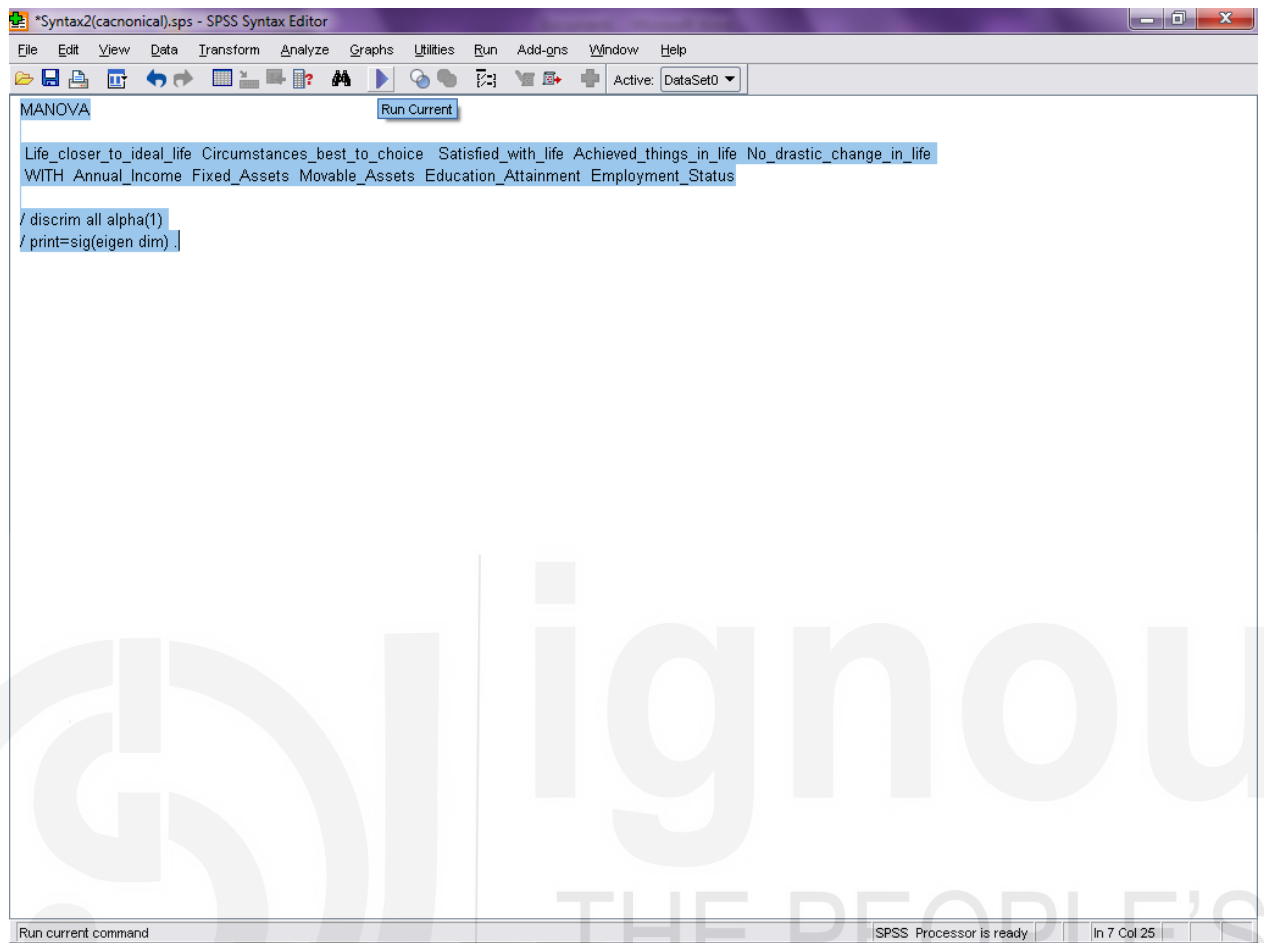


Step 2: Then type the following syntax :



Note that the criterion set of variables are listed before **WITH** and predictor variables are listed afterwards

Step 3: This command can be implemented using RUN button on toolbar menu.



The results of above illustration using SPSS has been presented in this section. The table 1 below shows an overall multivariate test of the entire model using different multivariate criteria.

Table 14.1: Multivariate Tests of Significance

Test Name	Value	Approx. F	Hypoth. DF	Error DF	Sig. of F
Pillais	.14258	2.40693	25.00	2050.00	.000
Hotellings	.15845	2.56302	25.00	2022.00	.000
Wilks	.86055	2.49153	25.00	1509.72	.000
Roys	.11708				.000

Table 14.2: Eigenvalues and Canonical Correlations

Root No.	Eigenvalue	Pct.	Cum. Pct.	Canon Cor.	Sq. Cor
1	.13260	83.69055	83.69055	.34217	.11708
2	.01720	10.85381	94.54437	.13003	.01691
3	.00698	4.40258	98.94695	.08323	.00693
4	.00167	1.05222	99.99917	.04080	.00166
5	.00000	.00083	100.00000	.00115	.00000

In this example, we are not only interested to know whether there is a relationship between the predictor and criterion variables but also wanted to know which indicators of economic well-being are more or less useful in explaining the relationship between life satisfaction and economic well-being. This is exactly where CCA comes into play. The interpretation and evaluation of the results has been discussed in next section.

14.7 INTERPRETATION OF CCA RESULTS

We now come to the conclusive stages of the Canonical Correlation Analysis. To draw the right conclusions amounts to the correct interpretation of the results obtained by conducting the CCA.

The first step is to evaluate the overall statistical significance of the full canonical model. This is done by testing the Null hypothesis. The Null hypothesis is that there is no relationship between the two set of variables. The alternative hypothesis is that the two sets of variables are related.

The interpretation of CCA is now accomplished by computing the F-statistic value using the Wilks- λ test.

We now come to the actual interpretation of the CCA as conducted in our example. Here, the value of the computed value of the F-statistic is high and significant. This is given in table 1 as 0.860. This means that we can reject the Null hypothesis. That is to say that we accept the alternative hypothesis. Thus, statistically significant relationship exists between life satisfaction and economic well-being indicators.

Let us recall, that the first root (function) is created in such a manner that the canonical correlation between the new variable is maximized and these new variables within each set are uncorrelated with zero mean and unit variance.

Let us interpret only those functions which explain reasonable amount of variance between variable sets. In our illustrations, we interpret only the first function as it explains 12% variance within the function as shown in table no. 2. All other functions explain less than 10% of variance in their functions, hence we can ignore them.

So, what we have concluded so far is that there is a statically significant relationship between our two set of variables. Further, this relationship is largely captured by the first root (function) in the canonical model. Next, we want to identify those variables which contribute significantly to explain this relationship between economic well-being and overall life satisfaction.

In multiple regression analysis, we often look at Beta weights to identify the relative contribution of one independent variable to explain dependent variable. In CCA, we look at the structure coefficient to decide which variables are useful for the model. Therefore, we examine the standardized weights and structure coefficients to interpret the first root (function). Let us underline the point that we are only concerned with the first function and will ignore other functions as they are not significant.

To understand the pattern among two set of variables, we have created table 3 showing coefficients which presents the standardized canonical function coefficients (i.e. weights) and structure coefficients for all variables. The squared structure coefficient (r_s^2) are also given, which represent the percentage of shared variance between observed variable and the new variable created from the observed variables set.

Table 14.3: Canonical Solution for Economic Well-being Indicator Predicting Overall Life Satisfaction for Function 1

Variable	Coef	r_s	r_s^2 (%)
Life closer to ideal life	.221	-.502	25.20
Circumstances of life best to my choice	-.692	-.905	81.90
Satisfied with life	-.223	-.729	53.14
Achieved things in life	-.400	-.746	55.65
Want no drastic change in life	-.054	-.427	18.23
Annual Income	-.381	-.857	73.44
Fixed Assets	.029	-.256	6.55
Movable Assets	-.210	-.696	48.44
Educational Attainment	-.668	-.895	80.10
Employment status	.219	-.286	8.17

Coef: Standized Canonical Function Coefficient, r_s = Structure Coefficient, r_s^2 = Squared Structure Coefficient

Looking at the coefficient, we find that relevant dependent variables are: the circumstances of life are best to my choice, I am satisfied with my life and I have achieved things in my life I aspired, because all these variables high squared structure coefficient which indicates the amount of variance the observed variable can contributed to new latent criterion variable. Looking at the other side of the equation on function 1 which involves predictor set, we find that Annual Income, Movable Assets and Educational Attainment variables were primary contributors. You must note that this process for interpretation of function is same as identifying the useful predictors in regression analysis with the exception that in canonical correlation analysis we have two set of equations for consideration.

Check Your Progress I

- 1) What do you mean by the term 'inter-disciplinary perspective'?

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- 2) How is CCA useful to address the inter-disciplinary nature of research questions?

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- 3) State the steps involved in computation of CCA through SPSS software.

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14.8 LIMITATIONS OF CANONICAL CORRELATION

- 1) The canonical correlation express the variance shared by the linear composites of the set of variables and not the variance extracted from the variables.